

VERKHNY UFALEY, Russian Federation

WMO: 285410

Lat: 56.083N

Lon: 60.300E

Elev: 1308

StdP: 14.01

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.1	-17.1	-28.2	1.2	-22.9	-22.3	1.7	-16.8	17.7	22.1	16.0	16.4	1.8	40	0.607

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.1	83.5	65.3	80.1	63.9	77.1	62.3	67.7	79.1	66.0	76.6	64.3	73.9	5.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.7	92.7	72.5	62.0	87.3	70.6	60.4	82.3	68.9	32.8	79.3	31.5	76.4	30.1	74.0	73.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.4	14.2	12.3	-29.4	88.3	6.2	3.2	-33.8	90.6	-37.5	92.5	-40.9	94.3	-45.4	96.6
			-29.6	70.1	6.2	1.6	-34.0	71.2	-37.6	72.2	-41.0	73.1	-45.5	74.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	35.7	7.1	10.2	23.0	38.0	50.7	59.0	62.6	59.1	48.8	36.7	21.0	10.7		
	DBStd	22.06	12.09	12.24	8.76	9.27	8.91	7.86	6.58	7.40	7.84	8.81	12.39	12.88		
	HDD50	6409	1331	1115	837	371	104	17	2	9	115	418	871	1219		
	HDD65	10804	1796	1535	1302	810	448	208	122	213	489	876	1321	1684		
	CDD50	1194	0	0	0	12	125	287	394	291	78	7	0	0		
	CDD65	115	0	0	0	0	5	29	49	30	2	0	0	0		
	CDH74	1406	0	0	0	4	108	374	541	343	36	0	0	0		
	CDH80	305	0	0	0	0	7	80	118	94	6	0	0	0		
(14) Wind	WSAvg	4.8	4.3	4.7	5.7	5.4	5.1	4.6	3.8	3.9	4.3	5.3	5.1	4.8		
(15) Precipitation	PrecAvg	22.1	1.1	0.8	1.1	1.4	2.1	2.8	3.6	2.7	1.7	2.0	1.6	1.3		
	PrecMax	31.1	2.4	1.9	2.0	3.7	5.4	5.3	8.1	5.3	4.1	4.7	2.9	2.7		
	PrecMin	14.5	0.2	0.2	0.2	0.1	0.4	0.7	1.3	0.6	0.3	0.2	0.2	0.3		
	PrecStd	4.3	0.5	0.4	0.6	0.9	1.3	1.2	1.7	1.3	0.8	1.1	0.7	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.4	36.7	49.3	71.5	80.7	86.4	87.9	87.8	78.7	64.7	46.6	34.9		
		MCWB	32.7	32.6	39.4	53.6	59.0	65.2	66.4	65.7	59.2	50.7	41.7	32.2		
	2%	DB	29.9	33.1	42.5	64.0	77.3	82.2	83.1	82.3	72.2	59.0	41.9	32.1		
		MCWB	28.3	29.8	35.5	48.7	57.7	64.5	66.0	65.6	57.7	48.6	39.2	30.2		
	5%	DB	27.0	30.2	38.3	58.6	73.3	78.4	79.8	77.4	66.6	54.0	38.2	30.3		
		MCWB	25.7	27.7	33.0	46.6	56.6	62.8	65.2	64.1	55.7	46.1	35.6	28.7		
	10%	DB	23.7	27.3	35.2	53.2	68.7	74.5	76.5	72.8	61.8	49.4	34.9	27.6		
		MCWB	22.2	24.9	31.3	43.6	53.9	61.2	63.6	62.4	53.4	44.3	33.0	25.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.1	33.8	40.2	54.1	63.2	68.9	69.7	69.1	63.0	53.0	42.5	32.6		
		MCDB	34.3	36.3	46.8	69.6	74.1	81.1	80.6	81.1	73.7	61.4	44.7	34.1		
	2%	WB	28.3	30.4	36.4	50.4	59.5	66.3	67.9	67.1	59.1	49.4	39.4	30.6		
		MCDB	29.6	32.5	41.0	60.8	72.9	77.9	79.0	78.6	69.4	56.8	41.8	32.0		
	5%	WB	25.8	27.8	33.9	47.4	57.4	64.2	66.4	65.0	56.6	47.0	35.8	28.8		
		MCDB	26.9	29.9	37.3	57.1	70.3	74.9	76.8	74.9	64.6	52.8	37.6	30.3		
	10%	WB	22.4	24.9	31.6	44.4	55.2	62.2	64.8	63.0	54.2	44.5	33.3	26.0		
		MCDB	23.6	27.3	34.8	52.5	66.2	72.3	74.2	71.4	60.8	49.1	34.7	27.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.1	16.4	16.5	18.0	21.6	19.8	19.1	17.8	16.2	11.2	10.2	11.5		
		MCDBR	10.5	16.1	20.5	27.3	29.0	26.6	26.0	26.5	24.5	22.3	12.2	10.2		
		MCWBR	9.6	13.2	14.6	14.6	13.4	11.6	11.2	11.7	12.5	13.0	9.9	9.3		
	5% WB	MCDBR	10.0	13.5	16.0	24.1	25.6	23.2	22.2	23.0	21.4	19.6	10.0	10.2		
		MCWBR	9.4	11.4	11.9	13.7	13.2	11.2	10.5	11.2	12.1	12.3	9.0	9.6		
(40) Clear-Sky Solar Irradiance	taub		0.277	0.304	0.359	0.402	0.402	0.396	0.427	0.437	0.378	0.345	0.302	0.274		
	taud		2.070	2.058	1.977	2.105	2.241	2.327	2.233	2.175	2.313	2.275	2.159	2.034		
	Ebn at Noon		189	233	246	256	266	269	256	243	244	219	179	154		
	Edh at Noon		20	30	42	43	40	37	40	40	30	24	18	16		
(44) All-Sky Solar Radiation	RadAvg		238	558	1015	1338	1607	1744	1698	1310	855	414	203	142		
	RadStd		25	43	78	151	167	163	133	134	66	80	26	23		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page